

The EEG normally remains relatively stable throughout adulthood, but it should be interpreted within the context of the physiological state. This chapter describes the normal EEG of adults during wakefulness, drowsiness, and sleep.

Wakefulness

In an awake resting adult, the normal EEG is characterized by a reactive, posterior-dominant alpha rhythm, usually in the range of 9–12 Hz with considerable individual variations. There is an anterior-predominant beta rhythm and admixtures of alpha and beta in the intermediate channels. Eye blink and muscle artifacts are present.

Additionally, certain normal variants such as lambda waves, posterior slow waves of youth and wicket waves may occur [1]. They are described in Chapter 13. Figure 8.1 shows a normal EEG during wakefulness.

Drowsiness

The typical transition to drowsiness is characterized by a decrease in the amplitude (attenuation) and slowing of the posterior-dominant alpha to theta temporally and occipitally. This change is especially prominent in the elderly. The anterior-predominant beta also increases in amplitude and slows to theta range, this leads to the emergence of diffuse theta (especially prominent in young adults). Slow lateral eye movements appear and muscle artifact dissipates. Figure 8.2 shows the normal EEG during drowsiness. Sometimes there may be a sudden symmetric sub-harmonic drop in frequency of the posterior-dominant rhythm (such as 10 Hz to 5 Hz). The point of the drop may resemble a sharp notch that can be mistaken for an abnormality. In others, the alpha may become paradoxically prominent with eye opening (paradoxical alpha). Rarely, beta enhancement also occurs [2]. Hypersynchrony may occur in young adults as described in Chapter 7.



Figure 8.1 Twenty-three-year-old man with epilepsy, normal wakefulness characterized by a reactive posterior-dominant rhythm (11 Hz), low-amplitude frontal beta activity, eye blink, and temporalis muscle artifact.

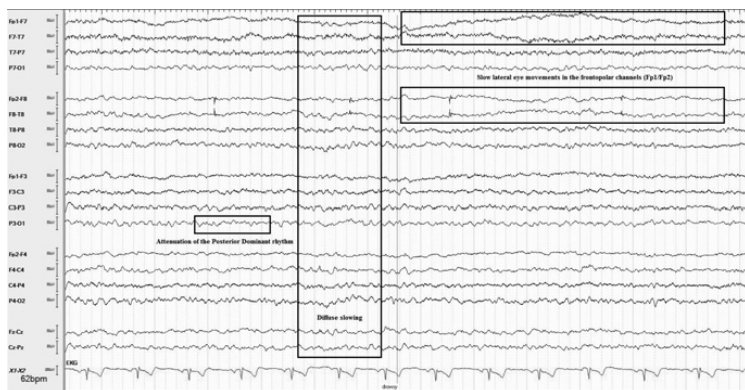


Figure 8.2 Same patient as in Figure 8.1, normal drowsiness characterized by attenuation of the posterior-dominant rhythm, emergence of theta range slowing and slow lateral eye movements. Note that the muscle artifact has dissipated.

Sleep

Normal sleep consists of non-rapid eye movement (NREM) and rapid eye movement (REM) sleep. Non-rapid eye movement sleep is further classified into stages I, II, and III (slow-wave sleep).

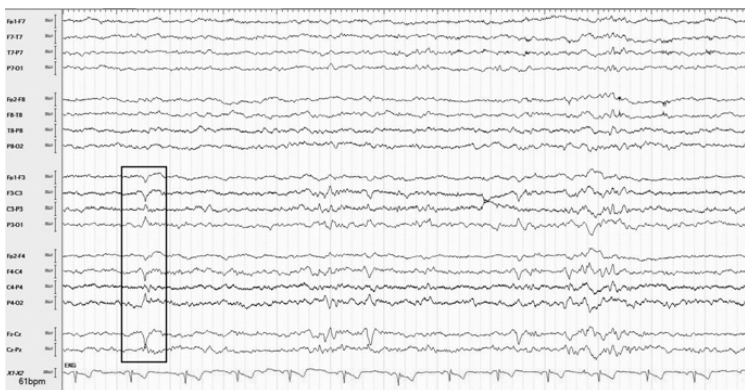


Figure 8.3 Same patient as in Figure 8.1, stage I sleep with vertex waves and POSTS.

Stage I (N1) sleep is marked by the appearance of vertex waves in the backdrop of drowsiness.

Vertex waves are centrally predominant (vertex), sharp transients (200 ms or more in duration) with maximal negativity at the Cz channel. They are characteristically bilateral, symmetric, and synchronous without an aftercoming slow wave. They may be induced by a sudden noise and abolished by arousal. In childhood and youth, they may have a prominent and sharper appearance but unlike central spikes (abnormality), vertex waves are state dependent and not consistently lateralized. Figure 8.3 shows vertex waves during stage I sleep.

Positive occipital sharp transients of sleep (POSTS) may also occur. These are described in Chapter 13.

Stage II (N2) sleep is marked by the appearance of sleep spindles and K complexes. They appear as cerebral activity progressively slows into theta and later delta range.

Sleep spindles are transient, frontocentral-predominant bursts of sinusoidal 10–14 Hz activity. They are normally symmetric and synchronous with maximal amplitude in the middle and tapering ends (spindle form).

K complex is a high-amplitude, central-predominant, diphasic complex with an initial sharp transient followed by a large slow wave with an associated sleep spindle that is best appreciated in the frontocentral region. K complexes may be induced by a loud noise. Persistent asymmetries of spindles and K complexes are suggestive of cerebral pathology ipsilateral to the side of their absence or attenuation. Figure 8.4 shows sleep spindles and K complexes during stage II sleep.

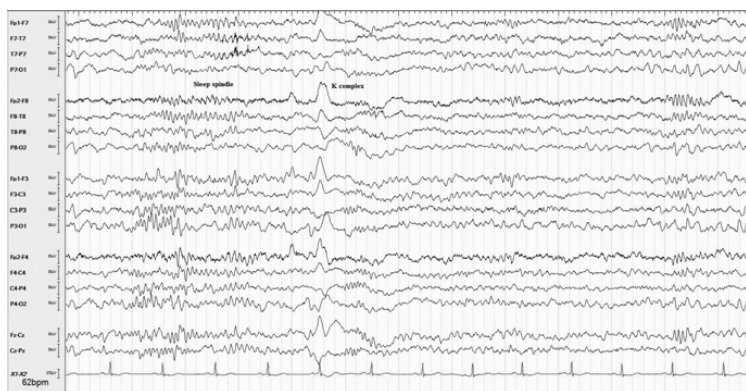


Figure 8.4 Same patient as in Figure 8.1, stage II sleep with sleep spindles and K complexes.

Sometimes, a sharper waveform may occur on the upslope of the slow wave of same polarity. This resembles the outline of a hand mitten with the “thumb” formed by the last wave of the spindle and the “hand” formed by the slow wave. These *mitten waves* are normal variants of the K complex. They have a frontocentral predominance and parasagittal spread. They are best appreciated on a referential montage. Figure 8.5 shows mitten waves. Rarely, epileptic discharges may be associated with K complexes as shown in Figure 8.6 – this is called *dyshormia* [3]. Epileptic discharges are described later in this book.

Stage III (N3 slow-wave sleep) is characterized by the emergence of higher amplitude (>75 μV), diffuse, 1–2 Hz semirhythmic delta slowing. Vertex waves, spindles, and K complexes persist during this stage. Figure 8.7 shows slow-wave sleep.

Rapid eye movement (REM sleep) is characterized by rapid eye movements, hypotonia (absence of muscle artifact) and sawtooth waves on the EEG. The background shows range slowing without any muscle artifact (like drowsiness). Rapid eye movements manifest as quick saccades in the frontopolar channels unlike the slow eye movements of drowsiness. Central sawtooth waves appear as rhythmic slowing in the central and parietal channels best seen on a referential montage. Figure 8.8 shows REM sleep. Most normal adults will spend 75% of their sleep in NREM and 25% in REM, these alternate in cycles four to six times in a normal night’s sleep, each cycle about 90–110 min in duration. There is usually a predominance of NREM in the first part of the night and REM in the later third. Occasionally, a routine EEG may capture REM – this is usually from sleep deprivation and doesn’t necessarily imply a diagnosis of narcolepsy [4].

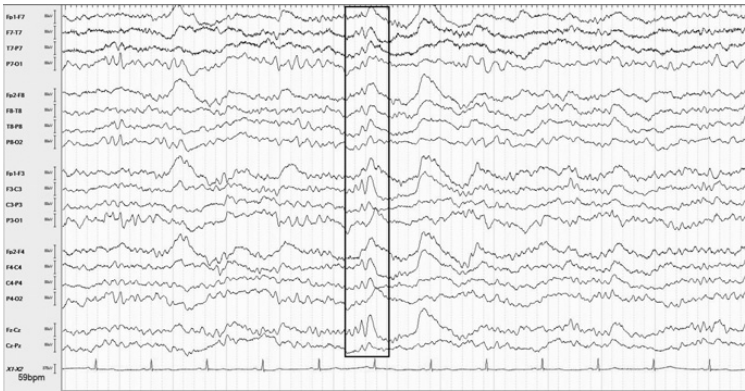


Figure 8.5 Same patient as in Figure 8.1, mitten waves.

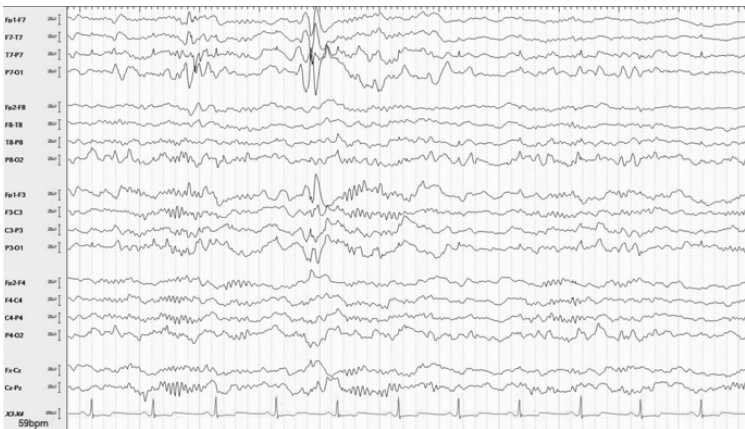


Figure 8.6 Fifty-two-year-old woman with mesial temporal lobe epilepsy, left temporal discharges associated with K complexes (dyshormia).

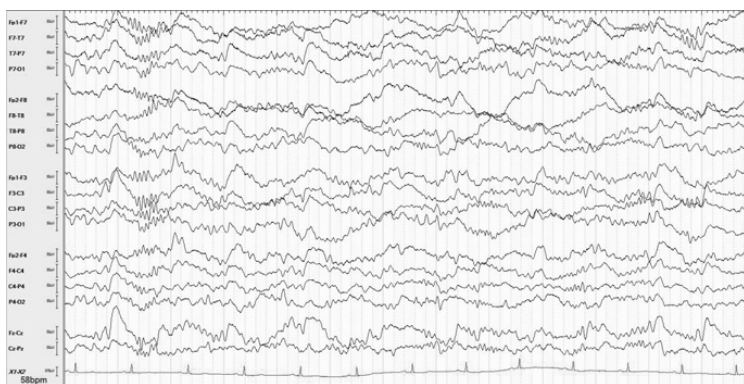


Figure 8.7 Sixty-two-year-old man with epilepsy, slow-wave sleep characterized by diffuse, high-amplitude, semirhythmic delta slowing.

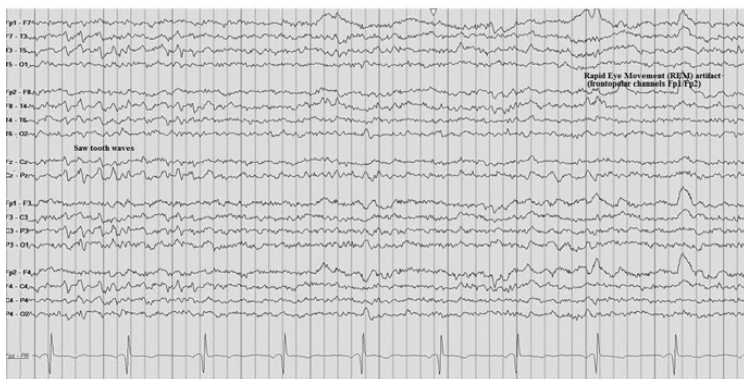


Figure 8.8 Thirty-two-year-old man with nocturnal movements, rapid eye movement artifact, and sawtooth waves during REM sleep.

Chapter Summary

1. The normal EEG doesn't change much during adult life, but it must be interpreted in the context of physiological states (awake, drowsy, or asleep).
2. Normal wakefulness is characterized by a reactive posterior-dominant alpha rhythm, anterior faster beta activity, eye blink, and muscle artifact.
3. Transition to drowsiness is typically characterized by attenuation of the posterior-dominant rhythm, diffuse slowing into theta range, emergence of slow lateral eye movements, and dissipation of muscle artifact.
4. Vertex waves are the architectural feature of stage I sleep. Positive occipital sharp transients of sleep (POSTS) may also occur.
5. Sleep spindles and K complexes are the architectural feature of stage II sleep.
6. Mitten waves are a normal variant, while dyshormia is abnormal.
7. Slow-wave sleep is characterized by diffuse high-amplitude semirhythmic delta slowing.
8. Rapid eye movement sleep is characterized by eye movement artifact and sawtooth waves.

References

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